Analysis and comparison

1 Introduction

Electroencephalography (EEG) provides a noninvasive measurement of brain electrical activity with high temporal resolution. Learning representations that transfer across tasks, participants, and recording conditions is a central objective for EEG foundation models. Recent approaches, including LaBraM, CBraMod, EEGPT, and REVE, use large-scale self-supervised pretraining to reduce dependence on task-specific annotations and support multiple downstream applications [5, 11, 10, 4]. Their evaluation also benefits from established supervised baselines such as EEGNet, a compact convolutional architecture rather than a pretrained foundation model [7].

Strong decoding performance, however, does not by itself establish which properties of a recording support a prediction. EEG representations may encode task-related neural activity alongside participant identity, session characteristics, acquisition settings, and physiological or instrumental artifacts. These factors can become predictive when they correlate with the target labels. Recent diagnostic work reports substantial subject-related structure in several foundation models, while broader evaluations reveal model-dependent robustness to noise and channel loss [8, 9]. Subject information is not necessarily an artifact: it may also reflect meaningful individual physiology. Nevertheless, these findings motivate examining what a representation preserves, beyond its downstream accuracy.

Our exploratory inspections of latent spaces across several EEG datasets motivate a closer investigation of this issue. In particular, the four-class motor-imagery task in BCI Competition IV dataset 2a provides a concrete setting in which to examine whether class separation reflects motor-related activity or correlated nuisance signals. The dataset includes electrooculographic recordings and expert artifact annotations, which can support controlled analyses [2]. Apparent separation in a latent-space visualization is insufficient to conclude that a model relies on artifacts, or that it fails to encode useful neural information. We therefore treat artifact dependence as a hypothesis to test through targeted controls and ablations, rather than as an established property of all EEG foundation models.

The pretraining objective is one possible source of differences in the information retained by an encoder. Existing methods do not all optimize the same reconstruction target: CBraMod and REVE reconstruct masked EEG inputs, LaBraM predicts discrete neural codes derived from a learned tokenizer, and EEGPT already incorporates alignment between learned representations [11, 4, 5, 10]. Reconstructing a waveform can encourage the preservation of any component that contributes to reconstruction error, including nuisance variation. This does not imply that signal-space learning is intrinsically uninformative, nor that a latent-space objective automatically removes artifacts. It instead raises the question of how the learning target and the construction of training views jointly influence the resulting representation.

Spatial organization provides a complementary inductive bias. Electrode locations constrain how measurements relate across the scalp, and recent models explicitly incorporate spatial structure through dedicated mixing mechanisms or positional information [11, 4]. At the same time, channel availability, electrode configuration, and reference choice alter the observations presented to a model. Montage effects must therefore be separated from differences in architecture and pretraining objective. Controlled montage ablations can help determine whether a model benefits from meaningful spatial relationships or remains dependent on a particular acquisition configuration. Incorporating electrode geometry alone does not establish invariance to these changes.

Joint-embedding learning offers a way to investigate an alternative to direct waveform reconstruction. LeJEPA combines agreement between views of the same sample with a regularizer that encourages an isotropic Gaussian embedding distribution, providing a mechanism to oppose representation collapse [1]. Its recent video extension, LeVJEPA, also investigates this principle for temporally structured data [6]. For EEG, views spanning different temporal contexts may encourage shared recording-level representations. However, their usefulness depends on which information the views share: persistent nuisance factors can also satisfy an agreement objective. The suitability of this approach for EEG therefore requires empirical evaluation.

We introduce EEGFM, a hierarchical encoder that combines electrode-wise tokenization, spatial mixing, bidirectional Mamba processing, and convolutional temporal modeling. Distance-dependent attention biases encode electrode geometry, and a learned-query bottleneck aggregates the encoded tokens into a global representation. We train this representation with a LeJEPA-inspired objective over global and local temporal views, without a waveform reconstruction loss. The study is organized around three objectives: characterizing task and nuisance information in existing representations, isolating montage effects through controlled ablations, and evaluating the contribution of spatial modeling and latent-space training in EEGFM.

2 Related Work

3 Methodology

We develop EEGFM, a hierarchical EEG encoder that combines electrode-wise signal tokenization, spatial mixing, and bidirectional temporal processing. The model produces both channel-resolved token embeddings and a global representation of each recording window. We train the global representation using a multiview objective inspired by LeJEPA [1], combining agreement between views with a distributional regularizer. The following formulation describes the patch-based configuration used in this work.

Let $X \in \mathbb{R}^{B\times C\times T}$ denote a batch of EEG windows, where B is the batch size, C the number of channels in a fixed canonical order, and T the number of time samples. Electrode coordinates are denoted by $P = (p_1, \dots, p_C)^\top \in \mathbb{R}^{C\times 3}$. A presence code $a_{bc} \in \{0, 1, 2\}$ accompanies each channel; code zero marks an absent channel, and both nonzero codes are treated as available by the attention masks. We write $m_{bc} = \mathbb{1}[a_{bc} \neq 0]$ and assume that each window contains at least one available channel. Hereafter, $\|$ denotes concatenation, $\odot$ element-wise multiplication, and RMSNorm and LayerNorm denote root-mean-square normalization and layer normalization, respectively. Features are column vectors, and linear maps act on their last dimension. The presence input is labeled p in the figures and a in the text.

3.1 Model Pipeline Overview

Figure 1 summarizes the mapping

$$(Z, g) = f_\theta(X, a; P), \tag{1}$$

where $Z \in \mathbb{R}^{B\times C\times N_4\times D_z}$ contains the token embeddings and $g \in \mathbb{R}^{B\times D_z}$ contains the global embeddings. These correspond to the outputs z and z_{glob} in the figures. The channel axis is preserved throughout the encoder; only temporal resolution and feature dimension change. The input length T is variable; the configured encoder uses $D_z = 384$. At a sampling rate of 200 Hz, an input window spans $T/200$ seconds.

Signal tokenization. The stem first applies a signed logarithmic compression,

$$\widetilde{X} = \operatorname{sign}(X) \odot \log(1 + |X|). \tag{2}$$

Each channel is processed independently by the same two temporal convolutional layers. Both have kernel size nine and are followed by eight-group GroupNorm and SiLU. The first convolution maps one input feature to $d_s = 32$ features; the second preserves this width and uses eight convolution groups. Padding preserves the time dimension. The resulting features are partitioned into non-overlapping patches of $p = 20$ samples. With $N_1 = \lfloor T/p \rfloor$, the flattened patch vector $r_{bcn} \in \mathbb{R}^{pd_s}$ is embedded as

$$H^{(0)}_{bcn} = W_{\text{stem}}\,\text{LayerNorm}(r_{bcn}) + e_c + e_{a_{bc}}, \tag{3}$$

where $W_{\text{stem}} \in \mathbb{R}^{128\times 640}$, e_c is a learned channel embedding, and $e_{a_{bc}}$ is a learned presence embedding. Any incomplete final patch is discarded.

Hierarchical encoding. The four stages use feature widths $(D_1, D_2, D_3, D_4) = (128, 192, 256, 384)$ and block counts $(L_1, L_2, L_3, L_4) = (2, 2, 3, 3)$. Between stages, adjacent tokens are concatenated, normalized, and projected:

$$r^{(i)}_{bcn} = H^{(i)}_{bc,2n-1} \| H^{(i)}_{bc,2n}, \tag{4}$$

$$H^{(i+1,\text{in})}_{bcn} = W^{(i)}_{\downarrow}\,\text{RMSNorm}(r^{(i)}_{bcn}), \tag{5}$$

where $W^{(i)}_{\downarrow} \in \mathbb{R}^{D_{i+1}\times 2D_i}$ and $n = 1, \dots, \lfloor N_i/2 \rfloor$. An unpaired final token is discarded. Thus, for patch size $p = 20$, the temporal lengths satisfy

$$N_i = \left\lfloor \frac{T}{p\,2^{i-1}} \right\rfloor, \qquad i = 1, \dots, 4. \tag{6}$$

In particular, $N_4 = \lfloor T/160 \rfloor$; we require $T \geq 160$ so that at least one temporal token reaches the bottleneck.

Token bottleneck and global readout. The final hidden tensor is normalized and optionally projected:

$$Z_{bcn} = W_z\,\text{RMSNorm}(H^{(4)}_{bcn}). \tag{7}$$

Here W_z is the identity because $D_z = D_4$; a learned projection is used when these dimensions differ. To obtain g_b, the CN_4 tokens are flattened into a sequence $(z_{bj})_j$. For each of $A = 8$ attention heads, linear maps produce keys and values of width $d_z = D_z/A = 48$:

$$k_{bhj} = \text{RMSNorm}(W^K_h z_{bj}), \qquad v_{bhj} = W^V_h z_{bj}. \tag{8}$$

Each head has $Q = 4$ learned queries $q_{hq} \in \mathbb{R}^{d_z}$. A token inherits the availability of its channel, denoted by m_{bj}. The pooled head outputs are

$$\alpha_{bhqj} = \frac{m_{bj}\exp(q^\top_{hq} k_{bhj}/\sqrt{d_z})}{\sum_r m_{br}\exp(q^\top_{hq} k_{bhr}/\sqrt{d_z})}, \tag{9}$$

$$o_{bhq} = \sum_j \alpha_{bhqj} v_{bhj}. \tag{10}$$

Concatenating the heads for each query and then the queries gives $o_b \in \mathbb{R}^{QD_z}$. The global representation is

$$g_b = W_{\text{out}} W_{\text{merge}}\,\text{RMSNorm}(o_b), \tag{11}$$

with $W_{\text{merge}} \in \mathbb{R}^{D_z\times QD_z}$ and $W_{\text{out}} \in \mathbb{R}^{D_z\times D_z}$. Thus, the global readout aggregates both channel and temporal information while excluding unavailable channels as attention keys. Training uses g_b directly; the token output Z remains available for downstream analysis.

3.2 Encoder Architecture

Every block preserves its input shape $B \times C \times N \times D$. As shown in Figure 2, its four residual updates take the form

$$U_r = U_{r-1} + \text{DP}_\rho\big(\gamma_r \odot F_r(\text{RMSNorm}_r(U_{r-1}))\big), \tag{12}$$

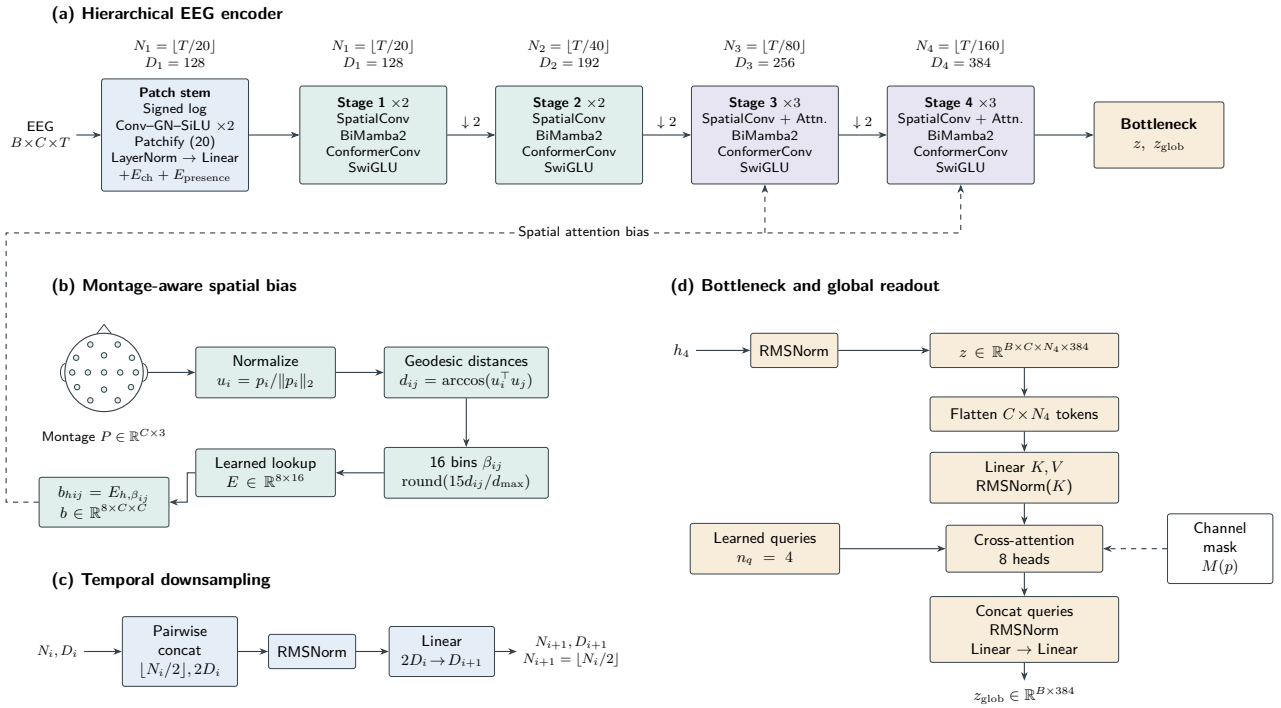


Figure 1: EEGFM architecture. A patch stem feeds four encoder stages with progressively shorter temporal sequences and larger feature dimensions. Geodesic distances between electrode coordinates are quantized into 16 bins and mapped to learned attention biases in the last two stages. The bottleneck returns token embeddings z and a global embedding z_{glob} obtained with four learned queries. The input has shape $B \times C \times T$; stage lengths are $N_i = \lfloor T/(20 \cdot 2^{i-1}) \rfloor$ for $i = 1, \ldots, 4$. Feature widths correspond to the configured encoder.

for $r = 1, \ldots, 4$, with F_r denoting spatial mixing, temporal Mamba2, Conformer convolution, and SwiGLU, respectively. Each branch has its own normalization parameters and feature-wise LayerScale vector $\gamma_r \in \mathbb{R}^D$, initialized to 10^{-4}. During training, DropPath multiplies a branch by $\xi/(1-\rho)$, with $\xi \sim \text{Bernoulli}(1-\rho)$ independently sampled per batch example and branch. The rate ρ increases linearly from zero to 0.15 over the ten blocks. Ordinary dropout is zero in the reference configuration.

Convolutional spatial mixing. Let U denote the input to the spatial module, already normalized by the outer residual branch. SpatialConv applies an additional RMSNorm and masks unavailable input channels:

$$G_{bcnd} = m_{bc}\,[\text{RMSNorm}(U_{bcn})]_d. \tag{13}$$

At every temporal and feature index, two 1×1 convolutions mix the channel axis:

$$S^{\text{conv}}_{b,:,n,d} = W_{c,2}\,\text{SiLU}(W_{c,1} G_{b,:,n,d}), \tag{14}$$

where $W_{c,1}, W_{c,2} \in \mathbb{R}^{C\times C}$. This branch is present in all stages and uses a fixed channel ordering.

Montage-dependent attention bias. Coordinates p_i are taken from the standard 10–20 montage for the mapped channel names. We normalize them and compute angular distances:

$$u_i = \frac{p_i}{\max(\|p_i\|_2, \varepsilon)}, \tag{15}$$

$$d_{ij} = \arccos\big(\text{clip}(u_i^\top u_j, -1, 1)\big)\,. \tag{16}$$

The distances are discretized into $K_d = 16$ bins,

$$\beta_{ij} = \text{round}\left(\frac{(K_d - 1) d_{ij}}{\max(\max_{r,s} d_{rs}, \varepsilon)}\right), \tag{17}$$

where the denominator is clamped from below by $\varepsilon = 10^{-8}$. The resulting integer indices are fixed buffers. A learned table $E \in \mathbb{R}^{A\times K_d}$, initialized to zero, defines the head-specific bias $b_{hij} = E_{h,\beta_{ij}}$. This table is shared across the attention blocks of the encoder.

Statistics-conditioned spatial attention. Stages three and four add a beamformer-style attention branch to Equation (14). For each channel, queries and keys are conditioned on statistics over the temporal tokens:

$$\mu_{bc} = \frac{1}{N}\sum_{n=1}^{N} U_{bcn}, \tag{18}$$

$$v_{bc} = \frac{1}{N}\sum_{n=1}^{N} (U_{bcn} - \mu_{bc})^{\odot 2}, \tag{19}$$

$$\ell_{bc} = \log\big(\max(v_{bc}, \varepsilon)\big), \tag{20}$$

$$s_{bc} = \text{LayerNorm}(\mu_{bc} \| \ell_{bc}). \tag{21}$$

The maximum and logarithm are element-wise, with $\varepsilon = 10^{-8}$. Each of the eight heads has feature width

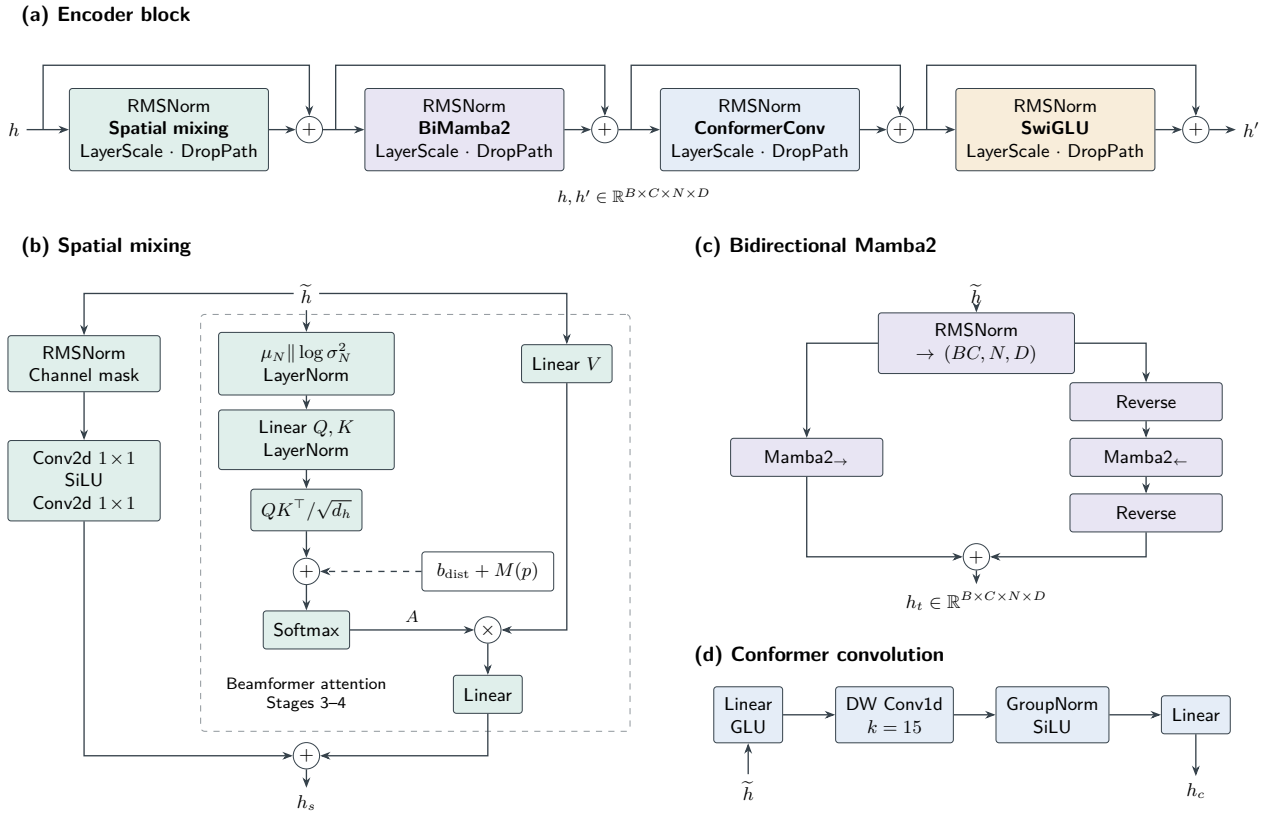


Figure 2: Encoder block and its constituent operations. Four pre-normalized residual branches perform spatial mixing, bidirectional Mamba2 processing, local temporal convolution, and a gated feed-forward transformation. Spatial attention is added to the convolutional spatial branch in stages three and four. $\widetilde{h}$ denotes a branch input after its outer normalization; $M(p)$ in the diagram is the additive channel-availability mask associated with the presence codes a in the text.

$d_h = D/A$ and computes

$$q_{bhi} = \mathrm{LayerNorm}(W_h^Q s_{bi}), \tag{22}$$

$$k_{bhj} = \mathrm{LayerNorm}(W_h^K s_{bj}). \tag{23}$$

The spatial weights are

$$e_{bhij} = q_{bhi}^\top k_{bhj}/\sqrt{d_h} + b_{hij}, \tag{24}$$

$$A_{bhij} = \frac{m_{bj}\exp(e_{bhij})}{\sum_r m_{br}\exp(e_{bhir})}. \tag{25}$$

Values retain the temporal axis: $v_{bhjn} = W_h^V U_{bjn}$. For each time index, the attention output is

$$S_{bin}^{\mathrm{attn}} = W_O \left(\Big\Vert_{h=1}^{A} \sum_j A_{bhij} v_{bhjn} \right). \tag{26}$$

A single spatial weight matrix per window and head is therefore reused across all temporal tokens. The spatial residual branch receives S^{conv} in stages one and two and $S^{\mathrm{conv}} + S^{\mathrm{attn}}$ in stages three and four. The geometric bias informs attention about relative electrode locations; the channel embeddings and convolutional mixing nevertheless retain a fixed channel indexing, so this construction does not imply arbitrary-montage or channel-permutation invariance.

Bidirectional temporal state-space processing. Temporal dependencies are modeled with two independently parameterized Mamba2 modules [3]. After an internal RMSNorm, the tensor is reshaped to (BC, N, D), treating each electrode as a separate sequence. Writing R for reversal along the token axis and $\bar{U} = \mathrm{RMSNorm}(U)$, the bidirectional output is

$$F_{\mathrm{temp}}(U) = \mathcal{M}_{\rightarrow}(\bar{U}) + R\big[\mathcal{M}_{\leftarrow}(R[\bar{U}])\big]. \tag{27}$$

Reversing the backward output restores the original token alignment before summation. Each Mamba2 module uses state dimension 128, expansion factor two, local convolution width four, and head dimension 64. This module mixes information across time within each electrode; interactions between electrodes are provided by the spatial branch.

For completeness, the selective state-space core of a single Mamba2 head can be expressed in recurrent form as

$$S_n = \eta_n S_{n-1} + b_n v_n^\top, \tag{28}$$

$$o_n = S_n^\top c_n + d\, v_n, \tag{29}$$

where $S_n \in \mathbb{R}^{128\times 64}$, $v_n \in \mathbb{R}^{64}$, and $b_n, c_n \in \mathbb{R}^{128}$. The scalar transition η_n and the input and readout vectors depend on the input; b_n includes the input

scaling from discretization, and d is a learned direct-path coefficient. This scalar-identity transition is the structured state-space component of Mamba2 [3]. The complete module also includes its input projections, local convolution, gating, normalization, and output projection. Equation (27) combines two such complete modules in opposite directions.

Local temporal convolution and gated feed-forward mapping. The Conformer branch supplements the state-space module with a local convolution. A linear map expands U from D to $4D$ features and splits it into $a, b \in \mathbb{R}^{2D}$. The gated features are $v = a \odot \sigma(b)$, where σ is the sigmoid. A depthwise temporal convolution of kernel size 15, with padding that preserves N, is followed by single-group GroupNorm, SiLU, and a projection back to D:

$$F_{\text{conv}}(U) = W_2 \,\text{SiLU}(\text{GN}(\text{DWConv}_{15}(v))) \,. \tag{30}$$

Finally, the feature-wise SwiGLU branch is

$$F_{\text{ffn}}(U) = W_3\big(\text{SiLU}(W_1 U) \odot W_2 U\big). \tag{31}$$

Its hidden width is $\lfloor 8D/3 \rfloor$ rounded upward to the nearest multiple of 64. All four operations are wrapped by the residual update in Equation (12).

3.3 Training Pipeline Overview

Multiview joint-embedding learning. LeJEPA combines a joint-embedding agreement objective with Sketched Isotropic Gaussian Regularization [1], abbreviated SIGReg. Our EEG adaptation applies both terms directly to the global readout in Equation (11). It uses a single shared encoder, with no separate teacher, stop-gradient target, predictor network, or additional projection head. The active training path does not reconstruct EEG signals.

For each source window X_b, the data loader supplies $V_g = 2$ global views and $V_\ell = 6$ local views. We denote their union by $x_b^{(v)} = \mathcal{T}_v(X_b)$, $v = 1, \ldots, V$, where $V = V_g + V_\ell = 8$ and the first V_g indices designate the global views. The augmentation configuration includes additive noise, global and channel-wise amplitude scaling, temporal shifts, and temporal, frequency, and spatial masking. The same presence codes a_b accompany all views of a window. Each view is passed through the same model:

$$y_b^{(v)} = g_\theta(x_b^{(v)}, a_b; P) \in \mathbb{R}^{D_z}. \tag{32}$$

Here g_θ denotes the global component of f_θ. Stacking these outputs yields tensors of shape $B \times V_g \times D_z$ and $B \times V_\ell \times D_z$, as shown in Figure 3.

Alignment with the global-view center. The target for every view of window b is the mean of its global embeddings,

$$c_b = \frac{1}{V_g} \sum_{v=1}^{V_g} y_b^{(v)}. \tag{33}$$

Both global and local embeddings contribute to the alignment loss,

$$\mathcal{L}_{\text{align}} = \frac{1}{BVD_z} \sum_{b=1}^{B} \sum_{v=1}^{V} \|y_b^{(v)} - c_b\|_2^2. \tag{34}$$

The factor D_z^{-1} matches averaging over features in the implementation. Gradients propagate through both the individual embeddings and the center c_b. Alignment alone admits a constant representation, motivating the additional distributional constraint.

Sketched isotropic Gaussian regularization. At each optimization step, we sample $K = 256$ unit directions $u_k = \omega_k / \|\omega_k\|_2$, with $\omega_k \sim \mathcal{N}(0, I_{D_z})$. For a fixed view v, the one-dimensional projections $s_{bvk} = u_k^\top y_b^{(v)}$ define an empirical characteristic function

$$\widehat{\varphi}_{vk}(t) = \frac{1}{B} \sum_{b=1}^{B} \exp(\mathrm{i} t s_{bvk}). \tag{35}$$

The target is the characteristic function of a standard normal variable, $\varphi_0(t) = \exp(-t^2/2)$. Define

$$e_{vk}(t) = |\widehat{\varphi}_{vk}(t) - \varphi_0(t)|^2 \varphi_0(t). \tag{36}$$

The implemented per-view regularizer approximates

$$\mathcal{R}_v = \frac{B}{K} \sum_{k=1}^{K} \int_{-5}^{5} e_{vk}(t)\, \mathrm{d}t \tag{37}$$

using the trapezoidal rule on $J = 17$ equally spaced points $t_j = -5 + 10j/(J-1)$, $j = 0, \ldots, J-1$:

$$\mathcal{R}_v \approx \frac{B\Delta t}{2K} \sum_{k=1}^{K} \sum_{j=0}^{J-2} \big[e_{vk}(t_j) + e_{vk}(t_{j+1})\big], \tag{38}$$

where $\Delta t = 10/(J-1)$. The complex discrepancy is evaluated through its real and imaginary parts,

$$|\widehat{\varphi} - \varphi_0|^2 = (\Re\widehat{\varphi} - \varphi_0)^2 + (\Im\widehat{\varphi})^2. \tag{39}$$

This operates on the unnormalized embeddings, so the unit-variance target also constrains their scale. The final regularizer is

$$\mathcal{L}_{\text{SIG}} = \frac{1}{V} \sum_{v=1}^{V} \mathcal{R}_v. \tag{40}$$

Thus, views are not pooled together as independent samples for SIGReg. The same random directions are reused across views within a step by seeding the sampler with the global step index. If distributed training is initialized, characteristic-function sums and sample counts are aggregated across workers; B in Equations (35), (37) and (38) then denotes the aggregated batch size. The regularizer is evaluated in float32 when embeddings are produced in reduced precision.

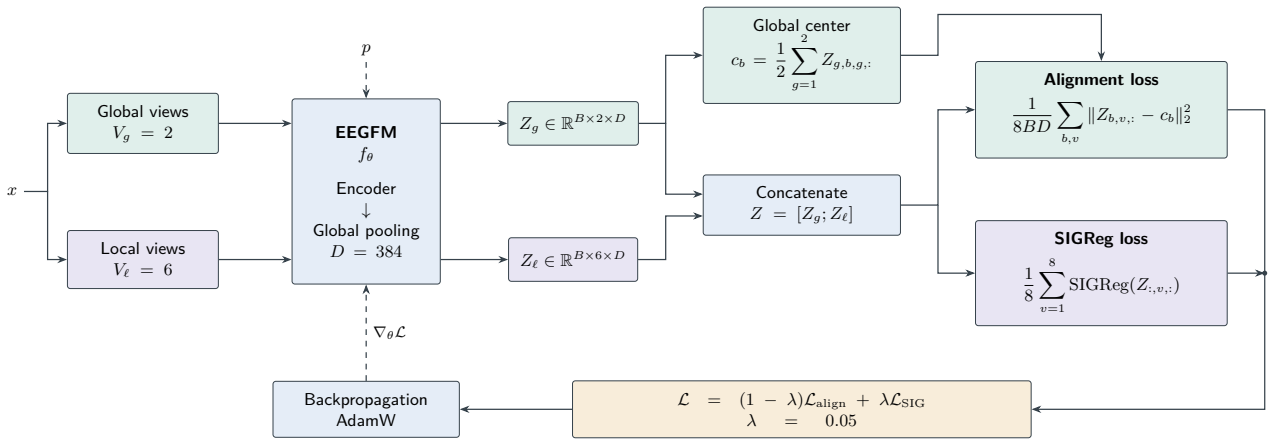


Figure 3: Multiview LeJEPA training. Two global and six local views are encoded by the same EEGFM parameters. All view embeddings are aligned with the mean of the global embeddings of their source window. SIGReg is evaluated across the batch separately for each view and then averaged. The weighted loss updates the complete encoder and global readout. The global center remains differentiable.

Objective and optimization. The complete objective is

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{align}} + \lambda\mathcal{L}_{\text{SIG}}, \qquad \lambda = 0.05. \tag{41}$$

The reference configuration selects TUEG windows with mapped channels and enables normalization in the data loader. The optimizer is AdamW with initial learning rate 10^{-4}, momentum coefficients $(0.9, 0.95)$, and weight decay 0.05 on the designated parameter group. Biases, normalization parameters, embeddings, learned queries, LayerScale vectors, and the spatial bias table are excluded from weight decay by the model's parameter-group rules. The total gradient norm is clipped at one. A cosine learning-rate schedule is stepped after every optimizer update, with a 10,000-step horizon and minimum learning rate 10^{-5}. The configuration uses batches of 16, bfloat16 autocasting, and gradient checkpointing within encoder blocks. It specifies 20 epochs, capped at 500 training batches and 100 validation batches per epoch.

The objective encourages agreement under the selected view transformations while penalizing departures from the Gaussian projection target. Neither term explicitly removes subject identity, acquisition-related information, or artifacts shared across views. Whether the resulting representations retain useful information under montage changes must therefore be established by downstream evaluations and ablations.

4 Results

4.1 Linear Probing

Linear probing assesses how readily task labels can be decoded from a pretrained representation without adapting the backbone. Each foundation model is frozen and used in evaluation mode to extract features $H = f_\theta(X)$. Since output shapes differ across models, all non-batch dimensions are flattened into one feature vector per example:

$$U = \mathrm{Flatten}_{\text{non-batch}}(H) \in \mathbb{R}^{B\times F}. \tag{42}$$

Thus, outputs of shape $B \times C \times D$ or $B \times C \times L \times D$ yield $F = CD$ or $F = CLD$, respectively, preserving channel and token features without pooling. Each backbone has its own probe and a fixed feature dimension within the evaluation setting. Only the linear classifier is trained on labeled training examples, using either a single affine layer with a classification loss or a linear support vector machine (SVM). Performance on held-out data measures the linear accessibility of task-related information in the frozen features. This limits the contribution of a flexible downstream classifier, while preserving the representations' full feature structure.

4.2 Ablation study on

5 Discussion and Conclusion

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